

Midterm study guide

Quantitative Reasoning II (INF 1220)

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1 Measurement and descriptive statistics

1.1 Operational definitions

An *operational definition* specifies the procedure by which a quantity is measured. The statistic reports only what that procedure covers. The US unemployment rate classifies a person as unemployed only if they are without work, currently available for work, and have actively searched within the preceding four weeks. A person who has stopped searching is classified as not in the labor force, so the unemployment rate excludes discouraged workers by construction. A change in a statistic can reflect a change in the quantity or a change in the definition.

1.2 Measures of center and spread

For observations $x_1, \dots, x_n$:

- **Mean:** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. Every observation enters with equal weight, so the mean is sensitive to extreme values.
- **Median:** the middle value of the ordered observations, or the mean of the two middle values when n is even. It depends only on the ranks of the observations, so it is resistant to extreme values.
- **Variance and standard deviation:** the sample variance is $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$, and the standard deviation s is its square root, in the units of the data. The population analogues are σ^2 and σ .

A *symmetric* distribution has its mean equal to its median. A gap between the two indicates *skew*, and the mean lies on the side of the longer tail: above the median in a right-skewed distribution, below it in a left-skewed one. Movie production budgets have a mean of \$31.1 million and a median of \$20.0 million, so the distribution is right-skewed and the median is the appropriate measure of a typical value.

1.3 Population and sample

The *population* is the complete set of units of interest. A *sample* is a subset of the population that is actually observed. A *simple random sample* gives every unit in the population an equal probability of selection. Inference from a sample to a population is valid when the sampling procedure makes the sample representative, which is a property of the procedure and not of the sample size. The Current Population Survey estimates the unemployment rate for roughly 275 million people from about 60,000 households by random sampling. The sampling assumption is stated again in Section 5.

2 Probability

2.1 Definitions and rules

The *sample space* S is the set of all possible outcomes of a random process. An *event* A is a subset of S . When the outcomes in S are equally likely,

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S}.$$

When probabilities are estimated from data, $P(A)$ is the relative frequency of A among the observed cases. A probability may also be a judgment with no counting behind it.

For events A and B :

- **Probabilities sum to one:** $0 \leq P(A) \leq 1$, and $P(S) = 1$. The probabilities of mutually exclusive and exhaustive events sum to 1.
- **Complement rule:** $P(\text{not } A) = 1 - P(A)$.
- **Addition rule for mutually exclusive events:** if A and B cannot both occur, then $P(A \text{ or } B) = P(A) + P(B)$.
- **Multiplication rule for independent events:** if A and B are independent, then $P(A \text{ and } B) = P(A) \times P(B)$.

Each of the last two rules holds only under its stated condition. Applying the addition rule to events that can both occur counts their intersection twice and overstates $P(A \text{ or } B)$. Applying the multiplication rule to dependent events understates $P(A \text{ and } B)$ when the events are positively associated, as when one flight delay makes a second more likely, and overstates it when they are negatively associated.

2.2 Conditional probability and independence

The *conditional probability* of A given B , for $P(B) > 0$, is

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)}.$$

It is the probability of A computed within the subset of outcomes in which B occurs. Events A and B are *independent* if and only if $P(A | B) = P(A)$, equivalently $P(A \text{ and } B) = P(A) \times P(B)$.

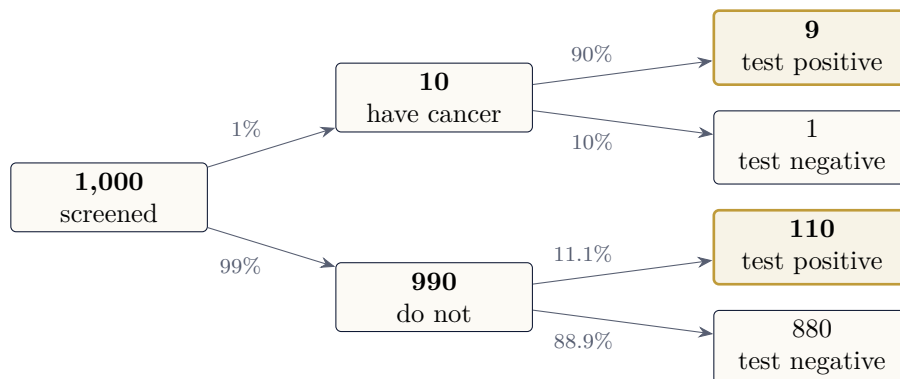
In general $P(A | B) \neq P(B | A)$. The two quantities share the numerator $P(A \text{ and } B)$ and differ in the denominator, so

$$\frac{P(A | B)}{P(B | A)} = \frac{P(A)}{P(B)}.$$

2.3 Base rates and screening

Let D be the event that a condition is present and $+$ the event that a test is positive. The test's *sensitivity* is $P(+ | D)$ and its *specificity* is $P(- | \text{not } D)$. The *base rate* or *prevalence* is $P(D)$. What a patient needs is the *positive predictive value*, $P(D | +)$, which depends on the base rate as well as on sensitivity and specificity. A probability tree over a reference population gives it directly.

For mammography with prevalence 10 per 1,000, sensitivity 0.90, and specificity 0.889 (rounded for the arithmetic; the BCSC figures for 2007–2013 are 0.869, 0.889, and about 6 per 1,000):



$$P(\text{cancer} \mid +) = \frac{9}{9 + 110} = \frac{9}{119} = 0.076.$$

Because the condition is rare, the 110 false positives from the 990 unaffected women outnumber the 9 true positives from the 10 affected women. Holding sensitivity and specificity fixed, the positive predictive value falls as the prevalence falls. In the language of updating, $P(D)$ is the *prior* probability and $P(D \mid +)$ is the *posterior* probability.

2.4 Expected value

For a discrete random variable X taking values x_i with probabilities $P(X = x_i)$,

$$E[X] = \sum_i x_i P(X = x_i).$$

The expected value is the probability-weighted mean of the possible values. By the law of large numbers it is the limit of the sample mean of repeated independent draws.

3 Random variables and distributions

A *random variable* is a numerical quantity whose value is determined by the outcome of a random process. Its *probability distribution* specifies the possible values and their probabilities. A distribution is described by its shape, its center, the mean μ , and its spread, the standard deviation σ .

Movie runtime is slightly skewed but roughly symmetric. The fair die is uniform: a *uniform* distribution assigns equal probability to every value. Production budget is heavily right-skewed. An *exponential* distribution describes waiting times between events in a process with a constant rate, and is right-skewed.

4 The normal distribution and the central limit theorem

4.1 The normal distribution

The *normal distribution* is a continuous, symmetric, bell-shaped distribution determined entirely by its mean μ and standard deviation σ .

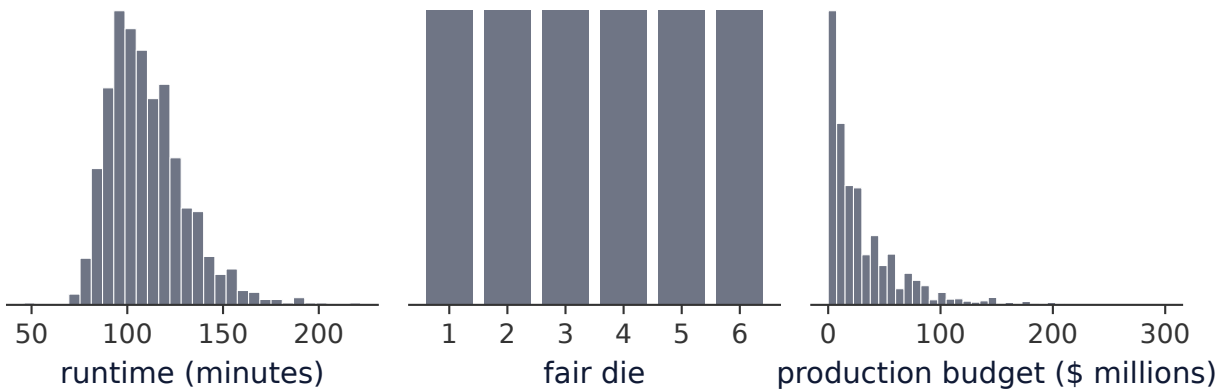


Figure 1: Two measured columns and one stipulated process. Movie runtime is slightly skewed but roughly symmetric; the fair die is uniform; production budget is heavily skewed.

The *standardized value* or *z-score* of an observation x is

$$z = \frac{x - \mu}{\sigma},$$

the number of standard deviations by which x lies above or below the mean. Standardizing places observations from different normal distributions on a common scale. For adult men's height, with $\mu = 68.3$ inches and $\sigma = 3.0$ inches, an observation of 74 inches has $z = 1.90$.

4.2 Normal distribution rule of thumb

For a normal distribution, approximately 68 percent of values lie within one standard deviation of the mean, 95 percent within two, and 99.7 percent within three. The exact coverages are:

Interval	Rule of thumb	Exact
$\mu \pm 1\sigma$	68%	68.27%
$\mu \pm 2\sigma$	95%	95.45%
$\mu \pm 3\sigma$	99.7%	99.73%

While we use 2σ as a rule of thumb, 1.96σ is the multiplier that gives an exact 95 percent confidence interval.

The rule is a property of the normal distribution and does not apply to other shapes. The following columns are real measured data. The first two roughly follow the normal distribution, while the second two do not and are heavily skewed:

	n	Mean	Median	SD	Within 1 / 2 / 3 SD
Height, inches	2,630	68.3	68.3	3.02	68.4 / 95.1 / 99.8
Movie runtime, minutes	1,209	110.2	107	20.2	70.6 / 95.8 / 98.8
Firm sales, \$M	4,700	4,255	367	20,355	96.9 / 98.3 / 98.8
Return on assets, %	4,820	-7.8	1.1	37.8	89.9 / 96.6 / 98.3

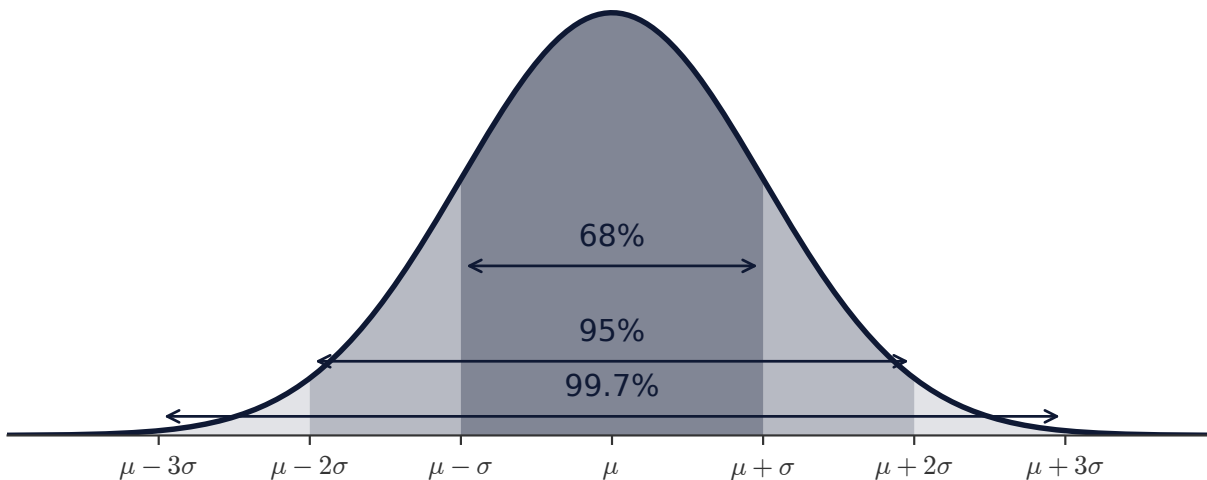


Figure 2: The normal distribution, with the shares of the distribution within one, two, and three standard deviations of the mean.

4.3 The sampling distribution of the mean

Let $X_1, \dots, X_n$ be independent draws from a population with mean μ and standard deviation σ , and let $\bar{X}$ be their mean. The *sampling distribution* of $\bar{X}$ is the distribution of $\bar{X}$ over repeated samples of size n . It has

$$E[\bar{X}] = \mu \quad \text{and} \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The standard deviation of the sampling distribution is the *standard error* of the mean.

4.4 The law of large numbers and the central limit theorem

The *law of large numbers* states that as n increases, $\bar{X}$ converges to μ . The precision of $\bar{X}$ as an estimate of μ therefore increases with n : its standard error $\sigma/\sqrt{n}$ falls toward zero. The law of large numbers does not say anything about the distribution of $\bar{X}$.

The *central limit theorem* states that as n increases, the sampling distribution of $\bar{X}$ approaches a normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$, regardless of the shape of the population distribution. It is what allows the normal distribution to be used for confidence intervals and hypothesis tests about μ .

The estimate of $\bar{X}$ improves with n , and the normal approximation is reached sooner for populations that are closer to symmetric. The mean of 25 draws from a uniform distribution is close to normal; the mean of 25 draws from a distribution as heavy-tailed as firm sales is still visibly skewed.

5 Confidence intervals and hypothesis testing

5.1 Standard error

The standard error of the sample mean is $\sigma/\sqrt{n}$. When σ is unknown it is estimated by

$$\text{SE} = \frac{s}{\sqrt{n}}.$$

The standard error decreases with the square root of the sample size: reducing it by half requires four times as many observations.

5.2 Confidence intervals

By the rule of thumb, 95 percent of the sampling distribution of $\bar{X}$ lies within two standard errors of μ , so

$$\bar{x} \pm 2 \times \text{SE}$$

is a 95 percent *confidence interval* for μ , with 1.96 in place of 2 for the exact figure. A sample of 100 women's heights with $\bar{x} = 62.73$ inches and $\text{SE} = 0.282$ inches gives the 95 percent interval [62.17, 63.29].

Over repeated samples, 95 percent of intervals constructed this way will contain μ . We can think of the 95 percent confidence interval as the set of reasonable values for μ given the sample.

5.3 Hypothesis testing and statistical significance

A hypothesis test weighs two hypotheses against the data. The *null hypothesis* H_0 is the default: the claim that there is no effect or no difference, taken as true unless the data provide evidence against it. The *alternative hypothesis* H_a is what the researcher suspects is true.

The *test statistic*

$$z = \frac{\bar{x} - \mu_0}{\text{SE}}$$

measures the distance of the sample mean from the value μ_0 that the null specifies, in standard errors. The *p-value* is the probability, computed assuming the null is true, of obtaining a test statistic at least as extreme as the one observed. The result is *statistically significant* at level α when $p < \alpha$; the conventional level is $\alpha = 0.05$, which corresponds to $|z| > 1.96$ in a two-sided test.

A confidence interval and a two-sided test at the same level are equivalent: the 95 percent confidence interval is the set of null values that would not be rejected at $\alpha = 0.05$.

5.4 Statistical versus practical significance

Statistical significance is a statement about $P(\text{data} \mid \text{null})$: how unlikely the observed data would be if the null hypothesis were true. It is not a statement about the size of the effect. Because the standard error shrinks with n , a difference of any size becomes statistically significant in a sufficiently large sample, so a result can be statistically significant and practically unimportant. Practical significance is a judgment about the magnitude of the estimate in the units of the problem.

5.5 The sampling assumption

Confidence intervals and hypothesis tests assume that the sample is a random sample from the population of interest. This assumption cannot be verified from the sample itself and is not corrected by a larger sample. A sample selected by a non-random mechanism, such as self-selection of respondents, supports inference only about the population that mechanism produces.

6 Simple linear regression

Regression is a statistical method for modeling the relationship between an outcome and one or more predictors. The outcome is the *dependent variable* Y and the predictors are the *independent variables* X . Regression is used for prediction and for explanation.

6.1 Correlation

The *correlation coefficient* r measures the strength and direction of the linear association between two variables, on a scale from -1 to $+1$. A value of $+1$ or -1 indicates an exact linear relationship; a value of 0 indicates no linear relationship. Correlation measures linear association only and can misrepresent a nonlinear relationship. Daily electricity use in Houston against the daily high temperature has $r = -0.533$ below the temperature at which cooling begins, $r = +0.915$ above it, and $r = +0.796$ over the full range, where the single coefficient summarizes a U-shaped relationship.

6.2 Simple linear regression

The simple linear regression model is

$$Y = b_0 + b_1X + e,$$

where b_0 is the *intercept*, in units of Y ; b_1 is the *slope*, in units of Y per unit of X ; and e is the *error* or *residual*. The *fitted value* $\hat{Y}_i = b_0 + b_1X_i$ is the model's estimate of $E[Y | X = X_i]$, the mean of Y among cases with that value of X , and the residual is $e_i = Y_i - \hat{Y}_i$.

In the Ames, Iowa data on 2,925 home sales, the regression of sale price on living area has slope \$116.23 per square foot and intercept \$6,773. A one-square-foot increase in living area is associated with a \$116.23 increase in sale price, on average. The intercept is the fitted price at zero square feet and has no substantive interpretation.

6.3 Least squares

The *least squares* estimates b_0 and b_1 are the values that minimize the sum of squared residuals (SSE), $\sum_{i=1}^n e_i^2$. Two properties of the least squares fit follow: the fitted values are perfectly correlated with X , and the residuals are uncorrelated with X .

6.4 R^2

The *coefficient of determination* R^2 is the proportion of the variance in Y accounted for by the regression, between 0 and 1. R^2 measures goodness of fit. It does not indicate whether the model is correctly specified or whether the relationship is causal.

6.5 Correlation and causation

A regression slope describes the association between X and Y in the data. It does not by itself establish that changing X would change Y .

7 Multiple linear regression

7.1 The multiple linear regression model

With k predictors the model is

$$Y = b_0 + b_1X_1 + \cdots + b_kX_k + e.$$

Each slope b_j is the average change in Y associated with a one-unit increase in X_j , holding all other predictors constant. The phrase “controlling for” refers to this holding constant. Least squares, fitted values, residuals, and R^2 are defined the same way as in simple linear regression. We still minimize the SSE, but minimize over a multidimensional space.

Adding a predictor never decreases R^2 and generally increases it, regardless of whether the predictor is informative.

7.2 Change in a coefficient when a predictor is added

A coefficient changes when a predictor is added if the new predictor is correlated with both the existing predictor and the outcome. In the Ames data, living area and quality rating have $r = 0.56$:

Outcome: sale price	Area only	Quality only	Both
Living area, per sq ft	\$116.23		\$62.78
Quality rating, per point		\$45,087	\$32,797
R^2	0.518	0.648	0.751

Area: holding quality constant, each additional square foot is associated with a \$62.78 higher price, on average.

Quality: holding area constant, each additional point of quality is associated with a \$32,797 higher price, on average.

The area-only coefficient of \$116.23 includes the part of the price difference that is attributable to the higher quality of larger houses. The two coefficients estimate different quantities: the total association between area and price, and the association between area and price among houses of equal quality.

7.3 Indicator variables

A *dummy variable*, also called an *indicator* or *binary variable*, takes the value 1 when a case belongs to a category and 0 otherwise. Its coefficient is the difference in mean Y between that category and the omitted *base category*, holding the other predictors constant. A categorical variable with k categories is represented by $k - 1$ indicators. Changing the base category changes every indicator coefficient without changing the fitted values. With one numeric predictor and one indicator, the model is two parallel lines, moved along the y -axis by the coefficient on the indicator variable.

8 Reading regression output

8.1 The columns

A regression table reports, for each predictor:

- **Coefficient:** the estimated slope b_j , in units of Y per unit of X_j , holding the other predictors constant.
- **Standard error:** the standard deviation of the sampling distribution of b_j .
- **t -statistic:** $t = b_j/\text{SE}(b_j)$, the number of standard errors by which the estimate differs from zero. $|t| > 1.96$ corresponds to significance at the 5 percent level.
- **p -value:** the p -value for the test of $H_0: b_j = 0$.
- **Confidence interval:** $b_j \pm 1.96 \times \text{SE}(b_j)$ at 95 percent. The interval contains zero when the coefficient is not significant at the 5 percent level.

Tables also report R^2 and the number of observations n .

8.2 Example

Sale price on five predictors in the Ames data. One house has no garage record and is dropped, so $n = 2,924$:

	Coefficient	Std. error	95% CI	p -value
Intercept	-66,415	5,159	(-76,531, -56,298)	< 0.001
Living area, sq ft	61.50	1.78	(58.01, 64.99)	< 0.001
Quality, 1-10	23,532	733	(22,095, 24,970)	< 0.001
Age at sale, years	-391	30	(-451, -331)	< 0.001
Central air, 0/1	2,378	2,927	(-3,361, 8,117)	0.42
Garage capacity, cars	13,444	1,211	(11,070, 15,818)	< 0.001
$R^2 = 0.784, \quad n = 2,924$				

Holding quality, age, central air, and garage capacity constant, each additional square foot is associated with a \$61.50 higher price, on average, with 95 percent confidence interval \$58.01 to \$64.99. Each additional year of age is associated with a \$391 lower price, and each additional car of garage capacity with a \$13,444 higher price.

The central air coefficient is \$2,378 with standard error \$2,927, confidence interval $-\$3,361$ to $\$8,117$, and $p = 0.42$. The coefficient is not statistically significant at the 5 percent level. This does not establish that central air has no association with price; the estimate is imprecise, and a non-significant result and a demonstrated absence of effect are different conclusions.

9 Controls and fixed effects

9.1 Controls and confounders

A *control* is a variable included in a regression so that it is held constant. Equivalent phrasings are “controlling for,” “adjusted for,” and “included as a covariate.” Which predictor is the control depends on the question.

A *confounder* is a variable associated with both the predictor of interest and the outcome. A confounder omitted from the model is a *lurking variable*, and the resulting distortion of the estimated coefficients is *omitted variable bias*. Including a measured confounder as a control removes its contribution from the coefficient of interest. An unmeasured confounder cannot be controlled for and leaves no indication in the output.

9.2 Fixed effects

Fixed effects are a set of indicator variables, one for each member of a grouping such as industry, state, or year. They are a control that holds constant every characteristic shared by the members of a group, including characteristics that are not measured. Equivalently, a model with group fixed effects estimates the slope from variation within groups only.

Consider gross margin, profit as a percent of sales after the cost of goods, and R&D share, research spending as a percent of sales, for 500 large US companies:

$$\widehat{\text{gross margin}} = 38.45 + 1.36 \times \text{R\&D share}, \quad R^2 = 0.35, \quad n = 500.$$

A one-percentage-point increase in R&D share is associated with a 1.36-percentage-point increase in gross margin. The standard error of the slope is 0.083, so $t = 16.3$ and the 95 percent confidence interval is [1.20, 1.52].

Research intensity varies systematically by industry, and the industries with the highest R&D share also have the highest gross margins. Five of the eleven industries in the sample:

Industry	Companies	Gross margin	R&D share
Healthcare	43	65.4%	13.4%
Business equipment	118	61.8%	13.3%
Manufacturing	55	33.7%	2.6%
Retail and wholesale	60	32.1%	0.2%
Utilities	36	35.2%	0.0%

Industry is therefore a confounder, and in the first model it is a lurking variable.

9.3 Fixed effects in a table and in output

Outcome: gross margin, %	(1)	(2)
R&D share, %	1.36	0.90
Industry fixed effects	No	Yes
R^2	0.35	0.47

Holding industry constant, a one-percentage-point increase in R&D share is associated with a 0.90-percentage-point increase in gross margin. The reduction from 1.36 to 0.90 is the part of the original association that was between industries rather than within them. Column (2) contains twelve estimated coefficients; published tables report the coefficient of interest and indicate the fixed effects with “Yes.”

Dep. variable: gross_margin	R-squared	0.468
Model: OLS	Adj. R-squared	0.456
Observations: 500	F(11, 488)	39.04

	coef	std err	t	P> t
const	49.7905	1.9238	25.88	0.000
R&D share	0.9029	0.0987	9.15	0.000
Healthcare	3.5547	2.7296	1.30	0.193
Manufacturing	-18.4042	2.7141	-6.78	0.000
Retail and wholesale	-17.8603	2.7492	-6.50	0.000
Utilities	-14.5652	3.1974	-4.56	0.000

The model contains ten industry indicators for eleven industries, of which five are shown. Business equipment is the base category, so each industry coefficient is the difference in intercept between that industry and business equipment, holding R&D share constant: 49.79 for business equipment, $49.79 + 3.55 = 53.34$ for healthcare, $49.79 - 17.86 = 31.93$ for retail. The slope on R&D share is 0.90 in every industry.

The healthcare coefficient has $t = 1.30$ and $p = 0.193$, so the difference in intercept between healthcare and business equipment is not statistically significant at the 5 percent level. R^2 rises from 0.35 to 0.468 with the addition of ten indicators, as R^2 always rises when predictors are added; the *adjusted* R^2 of 0.456 penalizes the number of predictors.

9.4 Limits of fixed effects

Group fixed effects hold constant characteristics that are common to a group and constant within it. They do not hold constant characteristics that vary between members of the same group, or that vary within a member over time. They address confounding at the group level only.

9.5 Causal inference

A causal effect is the difference between the outcome a unit experienced under one condition and the outcome the same unit would have experienced under another. The second outcome is never observed. A *randomized experiment* assigns units to conditions at random, so the groups are comparable in expectation on all characteristics, measured and unmeasured. A *natural experiment* is an observational setting in which assignment was determined by a process close to random. Without randomization, controls and fixed effects reduce confounding from measured and group-level sources but cannot rule out confounding from unmeasured sources, and a regression estimate should be reported together with what was held constant and what was not.